INCH-POUND

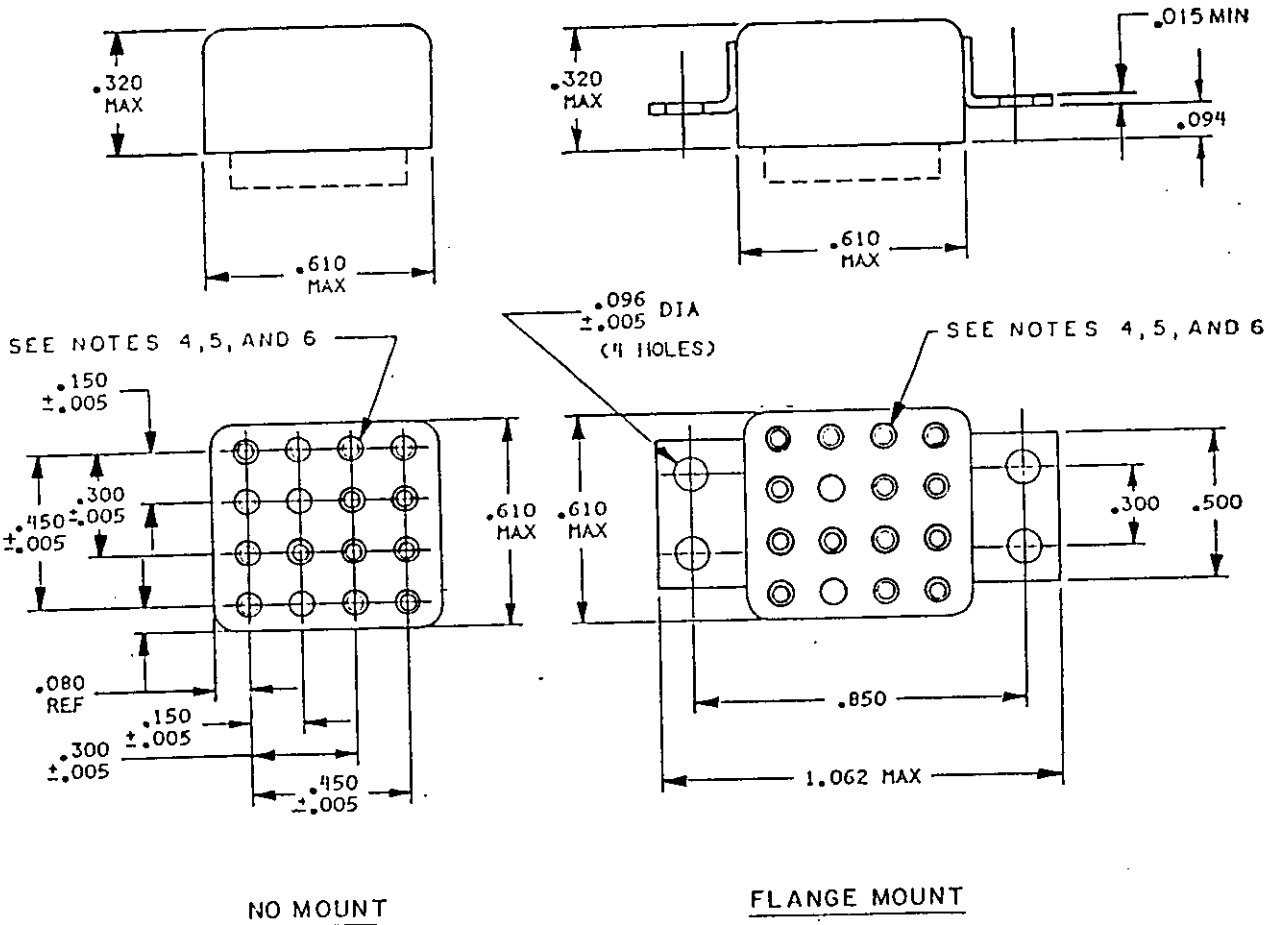
MIL-R-39016/653
24 August 1989

MILITARY SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, ESTABLISHED RELIABILITY, HERMETICALLY SEALED, 4PDT, LOW LEVEL TO 2 AMPERES (0.150-INCH TERMINAL SPACING) WITH INTERNAL DIODE FOR COIL TRANSIENT SUPPRESSION, INCLUDING INSPECTIONS OF APPENDIXES A AND B

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-R-39016.



NO MOUNT

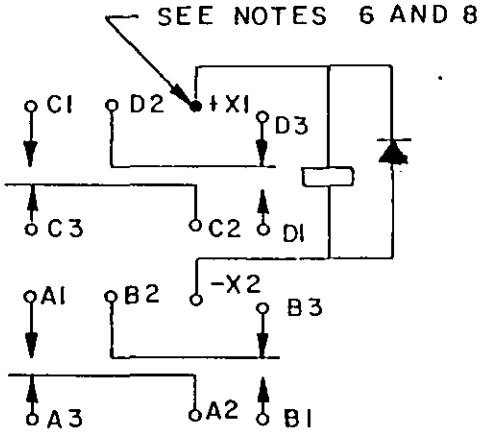
FLANGE MOUNT

FIGURE 1. Configurations and circuit diagrams.

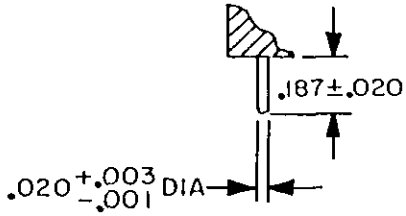
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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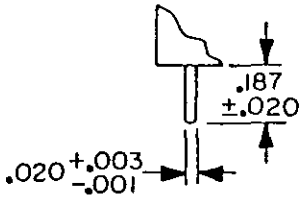
Inches	mm	Inches	mm
.001	0.03	.150	3.81
.003	0.08	.187	4.75
.005	0.13	.300	7.62
.015	0.38	.320	8.13
.020	0.51	.450	11.43
.050	1.27	.500	12.70
.080	2.03	.610	15.49
.094	2.39	.850	21.59
.096	2.44	1.062	26.97



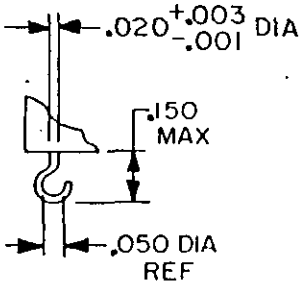
CIRCUIT DIAGRAM
(TERMINAL VIEW)



WIRE LEAD
PLUG IN (PI)
(SEE NOTE 9)



WIRE LEAD
SOLDER PIN - SP



SOLDER LUG
(SEE NOTE 7)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Terminal locating dimensions shown are applicable to all type mounts.
5. Terminal numbers in circuit diagram are for reference only. Numbers do not appear on relay.
6. Indicated terminal shall be identified with contrasting bead.
7. The shape of lug terminals is optional.
8. Coil symbol is optional in accordance with MIL-STD-1285.
9. Finish gold plated in accordance with MIL-G-45204, type II, class I, with a nickel underplating that shall be in accordance with QQ-N-290 (50 to 150 microinches thick), grade D except knoop hardness shall be 200 minimum.

FIGURE 1. Configurations and circuit diagrams - Continued.

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REQUIREMENTS:

Cleaning and small particle inspection: Applicable.

CONTACT DATA:

Load ratings:

High level (relay case grounded):

Resistive: 2 amperes at 28 V dc; 0.125 ampere at 115 V ac, (60 and 400 Hz); 0.5 ampere at 115 V ac, (60 and 400 Hz) with case not grounded. 400 Hz life test not required for qualification testing.

Inductive: 0.5 ampere at 200 mH inductive at 28 V dc.

Lamp: 0.10 ampere at 28 V dc.

Low level: 10 to 50 µA at 10 to 50 mV dc or peak ac.

Intermediate current: Applicable.

Contact resistance and voltage drop:

Initial: 0.050 ohm maximum.

High level:

During life: Not more than 5 percent of open circuit voltage.

After life: 0.150 ohm maximum.

Low level:

During life: 33 ohms maximum.

After life: 0.150 ohm maximum.

Intermediate current:

During intermediate current: 1 ohm maximum.

After intermediate current: 3 ohms maximum.

Contact bounce: 2 milliseconds (ms) maximum (applicable for after high level rated load life and intermediate current tests).

Contact stabilization time: 2.5 milliseconds (ms) maximum.

Overload (high level only): 4 amperes resistive at 28 V dc, 1.0 ampere inductive at 28 V dc (ac not applicable).

COIL DATA: See table I.

Operate time: 4.0 ms maximum over temperature range with rated coil voltage.

Release time: 6.0 ms maximum over temperature range from rated coil voltage.

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ELECTRICAL DATA:

Insulation resistance: 1/ 10,000 megohms minimum, except the resistance between coil and case at high temperature shall be 1,000 megohms minimum.

Dielectric withstanding voltage: 1/

	Sea level V rms (60 Hz)	Altitude V rms (60 Hz)
Between case, frame, or enclosure, and between all contacts in the energized and deenergized positions	750	350 All terminals to case
Between case, frame, or enclosure and coil(s)	500	
Between all contacts and coil(s)	750	
Between open contacts in the energized and deenergized positions	500	
Between contact poles	750	

DIODE CHARACTERISTICS:

Maximum negative transient: 1.0 V.

Coil transient suppression: Applicable (Warning: Reverse polarity on coil terminals will destroy the diode).

Semiconductor in process screening: Applicable, visual inspection of semiconductors shall be in accordance with MIL-STD-750, method 2074.

ENVIRONMENTAL DATA:

Temperature range: -65°C to +125°C.

Vibration (sinusodial): MIL-STD-202, method 204. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

Vibration (random): MIL-STD-202, method 214, test condition IG. Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contacts. Applicable to qualification, group C, and group A inspection testing.

Shock (specified pulse): MIL-STD-202, method 213, test condition C (100 g). Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

Magnetic interference: Applicable.

Resistance to soldering heat: Applicable.

Acceleration: Applicable.

Particle impact noise detection: Applicable.

1/ Connect coil leads together to avoid damage to diode.

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TABLE I. Dash numbers and characteristics. 1/

Dash number 2/	Mount	Leads	Coil voltage V dc 3/		At 25°C				Over temperature range		
			Rated	Max	Coil resistance Ohms ±10%	Specified pickup value voltage V dc	Specified hold value (voltage) (V dc)	Specified dropout value (voltage) (V dc)	Specified pickup value (voltage) (V dc)	Specified hold value (voltage) (V dc)	Specified drop-out value (voltage) (V dc)
001	Flange	Lug	6	7	28	2.7	1.6	0.3	3.8	2.2	0.18
002	No mount	Wire (SP)									
003	Flange	Wire (SP)									
004	Flange	Wire (PI)									
005	Flange	Lug	9	11	73	4.2	2.5	0.4	5.6	3.4	0.25
006	No mount	Wire (SP)									
007	Flange	Wire (SP)									
008	Flange	Wire (PI)									
009	Flange	Lug	12	14	115	5.4	3.2	0.6	7.6	4.3	0.36
010	No mount	Wire (SP)									
011	Flange	Wire (SP)									
012	Flange	Wire (PI)									
013	Flange	Lug	18	22	280	8.4	5.0	0.8	11.2	6.7	0.5
014	No mount	Wire (SP)									
015	Flange	Wire (SP)									
016	Flange	Wire (PI)									
017	Flange	Lug	22	27	430	10.3	6.0	1.0	14.0	8.0	0.6
018	No mount	Wire (SP)									
019	Flange	Wire (SP)									
020	Flange	Wire (PI)									
021	Flange	Lug	26.5	35	720	13.5	8.1	1.5	18.0	10.8	0.9
022	No mount	Wire (SP)									
023	Flange	Wire (SP)									
024	Flange	Wire (PI)									
025	Flange	Lug	36	42	1040	17.5	10.5	1.9	22.8	14.0	1.1
026	No mount	Wire (SP)									
027	Flange	Wire (SP)									
028	Flange	Wire (PI)									

1/ Each relay possesses high level and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 V dc maximum or peak ac open circuit are not recommended for subsequent use in low level applications.

2/ The suffix letter M, P, or R to designate the applicable failure rate level shall be added to the applicable listed dash number. Failure rate level (percent per 10,000 cycles): M, 1.0; P, 0.1; R, 0.01. Example, 001M - - - - - 003R.

3/ Caution: The use of any coil voltage less than the rated coil voltage will compromise the operation of the relay.

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PHYSICAL:

Terminals: See figure 1 and table I.

Terminal strength: 1.5 ±0.2 pounds (pull).

Solderability: Applicable (except to gold plated terminals).

Terminal twist test: Applicable to wire leads.

Dimensions and configurations: See figure 1 and table I.

Weight: 8.5 grams (0.30 ounce) maximum.

Seal: Hermetic, adjunct sealant shall not be used.

Identification marking (full): Applicable.

LIFE TEST REQUIREMENTS:

High level: 100,000 cycles.

Low level: 100,000 cycles plus 900,000 cycles mechanical life.

Part or Identifying Number (PIN): M39016/653-(dash number from table I and suffix letter designating failure rate level).

QUALIFICATION INSPECTION:

Qualification inspection and sample size: See table II.

Failure rate qualification: The manufacturer shall qualify his products to failure rate level "M" initially.

TABLE II. Qualification inspection and sample size. 1/

Single submission	Group submission	
33 units plus 1 open unit for level M at C = 0 2/ Qualification inspection as applicable.	M39016/653-021	33 units plus 1 open unit for level M at C = 0 2/ Qualification inspection as applicable.
	M39016/653-001 M39016/653-005 M39016/653-013 M39016/653-017 M39016/653-025 M39016/653-010 M39016/653-024	2 units, each PIN, qualification inspection group I, group II 2 units, qualification inspection table, group I, group II, and shock, vibration, acceleration, terminal strength, and seal.

See footnotes on next page.

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1/ For retention of qualification or extension of qualification to lower failure rate levels, all life test data accumulated on (or specified in) MIL-R-39016/53 and MIL-R-39016/654 may be used in addition to MIL-R-39016/653. Prior to performance of retention of qualification inspection testing, the relay manufacturer shall preselect the sampling plan.

2/ The number of units required for qualification testing will be increased as required in group V, table II, MIL-R-39016, if the contractor elects to test the number of units permitting one or more failures. Prior to the performance of qualification inspection testing, the relay manufacturer shall preselect the sampling size.

Qualification inspection (reduced testing) and sample size: If the relays produced for MIL-R-39016/653 are similiar in construction and design, except for the additional inspections specified herein, to the relays produced for MIL-R-39016/53, then reduced testing may be performed concurrent with or subsequent to successful qualification of MIL-R-39016/53 at the "M" failure rate level. See table III.

TABLE III. Qualification inspection (reduced testing) and sample size.

Examination or test
2 units each coil voltage, group A inspection table. 1 unsealed sample unit for internal inspection.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR
Navy - AS, OS, SH
Air Force - 99
DLA - ES

User activities:
Navy - MC
Air Force - 11, 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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